Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/16/2004
PI Name:	Javanainen, Juha D.Sc.		
Project Title:	Bose-Einstein Condensate and Atom Laser: Coherence and Optical Properties		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ji@phys.uconn.edu	**Fax:**	FY 860-486-3346
PI Organization Type:	UNIVERSITY	**Phone:**	860-486-3458
Organization Name:	University of Connecticut		
PI Address 1:	Department of Physics, U-3046		
PI Address 2:			
PI Web Page:			
City:	Storrs	**State:**	CT
Zip Code:	06269-3046	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/01/1998	**End Date:**	01/31/2003
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	6	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	End date should be 1/31/2003 per notes in Progress section (jp 2/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We are currently pursuing two main lines of research. First, we are developing a quantum field theory for photoassociation of a Bose-Einstein condensate and a degenerate Fermi gas. Second, we are studying the dynamics of ultralow-temperature gases in optical lattices. The objective is to generate new theoretical concepts in the physics of degenerate atomic gases, such as a Bose-Einstein condensate in an alkali vapor. When the concepts and ideas arising from this work circulate in the community, new theoretical ideas, experiments, and ultimately possibly new technologies will evolve. For instance, Feshbach resonance in a Fermi gas has already lead to a fermion superfluid, whereas a Bose-Einstein condensate in an optical lattice could become an enabling technology or		

	even the computational element in a quantum computer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The grant expired 01/31/03. All relevant information for the year 2003 is in the 2003 taskbook.
Bibliography Type:	Description: (Last Updated: 01/08/2020)